

Leon Meng

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EDUCATION

University of California, Irvine

Bachelor of Computer Science and Engineering

Irvine, CA

2025 – 2029

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, HTML, CSS, SQL

Web & Backend: React, Vite, FastAPI, Supabase, PostgreSQL, Vercel serverless functions, REST APIs

AI & Automation: Anthropic Claude & OpenAI APIs, AI image processing pipelines, LLM tooling, workflow automation

Robotics & Vision: Autonomous systems, odometry & localization, path planning (Roadrunner), finite state machines, OpenCV, Limelight 3A vision, CAD (Fusion 360), 3D printing

Developer Tools: Git, GitHub, Docker, Linux, CI/CD pipelines

EXPERIENCE

SomaFuture

Software Engineer

Oeiras, Portugal

Jun 2026 – Aug 2026

- Designed and built a completely new company website ([live demo](#)) that redefined SomaFuture's strategic direction toward standardizing and digitalizing its product line, translating a physical catalog into a structured digital experience
- Built and integrated AI and automation tools into internal workflows, streamlining repetitive tasks and accelerating the team's design and content production
- Prototyped 3D room configurators and evaluated outsourcing platforms against catalog-import, modular-configuration, and EU/GDPR requirements, authoring the comparative assessment used to guide the platform decision

First Tech Challenge — Teams 16031 & 16205

Autonomous & Mechanical Engineer

Vancouver, BC

Oct 2021 – Jun 2025

- Implemented an odometry system using encoders; computed horizontal/vertical displacement per time step, corrected for rotational drift, and converted to field coordinates via trigonometric transforms for sub-centimeter localization accuracy
- Built a vision-guided object detection system using Limelight 3A and OpenCV: designed a color-isolation pipeline (contour detection, color filtering) to detect, classify, and localize game elements in real time, then computed target contour angle and position to autonomously align the robot for pickup
- Implemented a finite state machine to coordinate all robot mechanisms via event triggers (button presses, timers), and integrated Roadrunner for trajectory planning and tuning to produce smooth, repeatable multi-step autonomous routines
- Designed and fabricated custom 3D-printed mechanical components (chassis, drivetrain, arm pivot, scoring mechanism) and co-authored the team's Engineering Portfolio; earned multiple top-scoring awards at British Columbia and Alberta championships

PROJECTS

GARDEROBE

Full-Stack Wardrobe Manager — React, Supabase, FastAPI, Claude API

the-garderober.com

- Built and shipped to production a full-stack wardrobe management app with AI auto-tagging that extracts item name, brand, color, and type from an uploaded photo via the Anthropic Claude API
- Implemented in-browser ML background removal, a drag-and-drop outfit builder, and a live price-tracking wishlist backed by scheduled FastAPI scraping jobs (Grailed, SSENSE) refreshed every 6 hours
- Built a fully featured social platform on Supabase realtime spanning all mainstream features — user profiles, publishing, an Explore page, a people directory, friends, and likes — plus a feed aggregating up-to-date fashion and culture articles from multiple third-party news APIs

BugLens

VS Code Extension — TypeScript, OpenAI & Anthropic APIs

github.com/ggttlplp201/BugLens

- Built a VS Code extension that explains the logic behind a highlighted bug — what the code does, what the user intended, and the underlying concept — without handing back a rewritten fix, to reinforce learning
- Integrated streaming LLM output across both OpenAI and Anthropic providers, bundled with esbuild and rendered in a WebView side panel

AWARDS & ACHIEVEMENTS

- USACO Gold Division (2023)